CHAINSTATE

A Symbolic-Weight Blockchain for Cognitive Transactions: Reputation-Weighted Bayesian Consensus over Distributed Language-Model Swarms with Post-Quantum Security and NWO-ASM / NEURO Composition

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Abstract

We present **CHAINSTATE**, a symbolic-weight blockchain whose unit of transaction is a cognitive query and whose unit of agreement is a reputation-weighted Bayesian log-pool over a distributed language-model swarm. The chain's weights are universal symbols — mathematical operators, scientific glyphs, full natural-language alphabets, alchemical sigils, Unicode 15.1 emoji and control-flow arrows — embedded in a 65,536-dimensional space across six structured subspaces with a learned cross-subspace interaction mask. Each transaction is dispatched to k ≥ 10 inference nodes; their per-node symbolic states are pooled through k-of-N log-pooling until cosine-agreement exceeds 0.95 (typically 3–7 rounds). Reputation evolves under an exponential-moving-average rule with stake-bounded cap; block production rotates through a reputation-weighted VRF on a 2-second target time, 64 transactions per block. The whole stack is wrapped by a post-quantum security envelope: CRYSTALS-Dilithium signatures (NIST FIPS 204), Kyber-1024 KEM (FIPS 203), and SHA3-256 commitments. CHAINSTATE composes natively with **NWO-ASM** for substrate-agnostic dispatch (GPU, photonic, neuromorphic, IBM and Origin quantum substrates) via Process-Matrix IR, and with **NWO NEURO** for live Mental State Signature conditioning of queries. We derive the math of the system, prove convergence and slashing-bound sybil resistance, compare CHAINSTATE on five axes against Bitcoin (PoW), Ethereum (PoS), Solana (PoH), Bittensor (AI-PoS), Avalanche and Polygon, and develop the philosophical position: CHAINSTATE makes cognitive work the unit of useful computation, makes belief aggregation primitive at the protocol layer, and offers a sovereign biometric-identity path through NWO NEURO and NWO Cardiac that is categorically incompatible with the regulatory direction of state-issued digital identity. A complete tech stack — Cloudflare Worker edge, PyTorch reference modules, ERC-1155 marketplace, USDC settlement on Base mainnet 8453 — ships open source as a static HuggingFace Space and a single-file Worker, both deployable in under five minutes by an individual developer.

Keywords: symbolic-weight blockchain · cognitive transactions · Bayesian log-pooling · proof-of-cognitive-work · post-quantum cryptography · NWO-ASM · NWO NEURO · open-source AI infrastructure · sovereign identity.

1. Introduction

The dominant deployment pattern for both blockchains and large language models is structurally identical: a small number of centralised actors operate the consensus or the inference, users access through a thin API, and the question of *what gets agreed on* is decoupled from the question of *what is useful to compute*. Bitcoin's proof-of-work secures a ledger by burning energy on cryptographic puzzles whose only output is the puzzle solution itself [1]. Proof-of-stake chains replace the puzzle with capital, but the energy expended on validation still produces no side-product the user wanted [2]. Centralised language-model APIs — OpenAI, Anthropic, Google — concentrate inference behind a few inference clusters;

the model is a single point of belief, a single point of failure, and a single point of capture. The pattern reaches a logical extreme in recent regulatory developments: the United Kingdom's Online Safety Act now mandates age-verification for swathes of the open Internet [3]; the European Union Digital Identity Wallet [4] reframes citizenship as a biometric token; and several jurisdictions tie native-application access to government-issued digital identity. The trajectory of state-controlled digital identity and the trajectory of centralised AI inference are, structurally, the same trajectory: gate computation, gate access, gate belief.

The present paper develops **CHAINSTATE**: a blockchain in which every transaction is itself a cognitive query, every block is a batch of resolved queries, and consensus is the reputation-weighted Bayesian posterior over a distributed swarm of inference nodes. The foundational mathematics — a thermodynamic reading of inference, a master equation for belief flow across a beacon graph, a log-pooling consensus rule, and a quorum-survival analysis showing robustness to large fractions of node loss — was developed in our earlier work on edge-resident language-model networks [5], and we treat that work as the foundational reference for the present paper. CHAINSTATE is the ledger-class instantiation: the same swarm, the same consensus, the same thermodynamic accounting, but exposed as a blockchain primitive with native settlement, native staking, native slashing and native marketplace economics on Base mainnet 8453.

Three commitments distinguish CHAINSTATE from existing blockchains. **(i) Cognitive work is the unit of consensus.** A node earns reputation by doing inference that the user paid for; the proof-of-cognitive-work compute proof is a SHA3-256 over (node id, query, timestamp, top-1024 dimensions of the emitted state), cheap to verify and expensive to forge without actually running the model at scale. **(ii) Universal symbols are the substrate.** Mathematics, science, language, occult, emoji and control-flow share one 65,536-dimensional embedding space with a learned cross-subspace interaction mask. A node that knows physics also knows chemistry; a node that knows astrology also knows the topology of state machines, because the embedding lets information flow across domains the way an educated human's intuition does. **(iii) The composition graph is the moat.** CHAINSTATE is layer-6 of a stack that includes NWO-ASM (Process-Matrix IR for substrate-agnostic dispatch — GPU, photonic, neuromorphic, IBM Sherbrooke quantum, Origin Wukong quantum, BCI, robotic) and NWO NEURO (live Mental State Signature conditioning of queries, signed under Dilithium FIPS 204). The economic graph is the same wallet, the same USDC settlement, the same affiliate splitter across all NWO services.

Structure. Section 2 fixes notation and lays out the ten-layer architecture. Section 3 develops the universal semiotic embedding and the symbolic-attention mechanism. Section 4 develops the consensus protocol with a convergence theorem. Section 5 covers block production, gas economics and staking. Section 6 specifies the post-quantum security envelope. Section 7 covers the NWO-ASM and NWO NEURO compositions. Section 8 contrasts CHAINSTATE with seven incumbent chains along five axes. Section 9 documents the developer surface, with worked code. Section 10 develops the political-economic position. Section 11 enumerates the verticals the protocol can serve. Section 12 is the roadmap, Section 13 the open theoretical questions, and Section 14 a brief conclusion.

2. Architecture

CHAINSTATE is a ten-layer stack. We label the layers L0 through L9 from the user-facing wallet at the bottom up to the live-conditioning bridge with NWO NEURO. Every layer is wrapped, end-to-end, by a single post-quantum security envelope: CRYSTALS-Dilithium signatures (NIST FIPS 204) [6], Kyber-1024 KEM (NIST FIPS 203) [7] and SHA3-256 commitments [8].

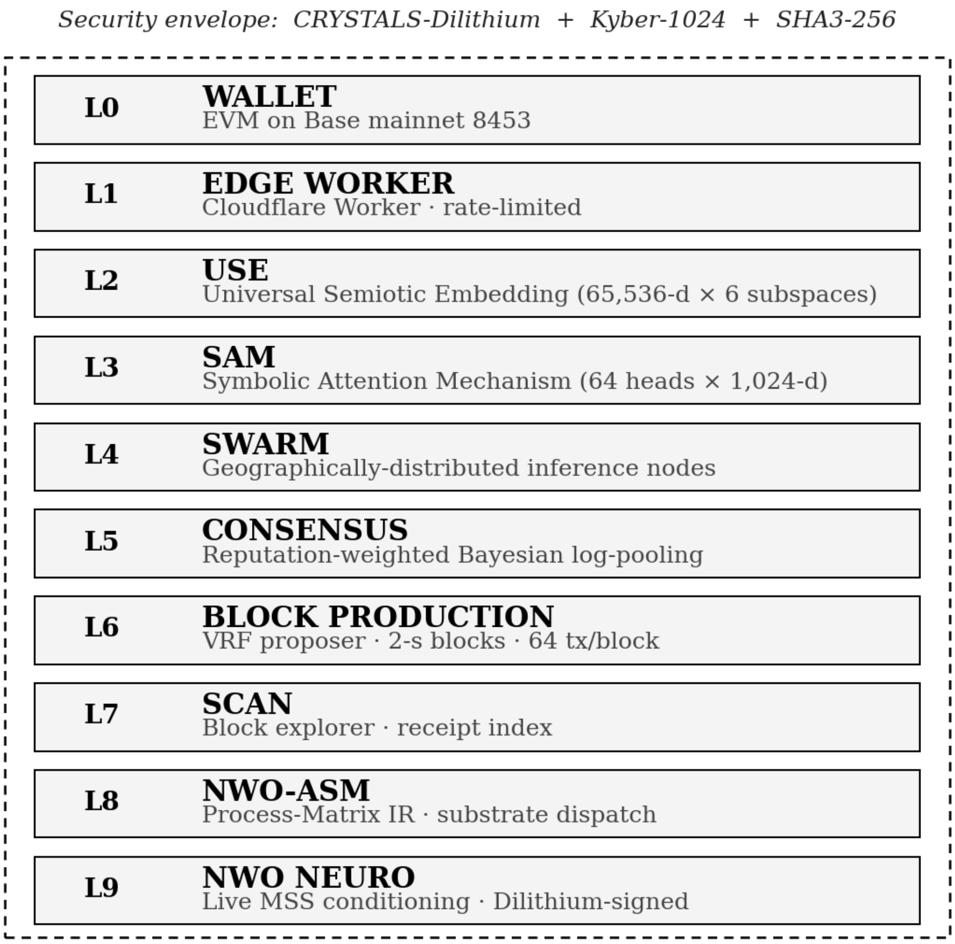

Figure 1. *The ten-layer CHAINSTATE stack. L0–L7 are owned by the protocol; L8 (NWO-ASM) and L9 (NWO NEURO) are sibling services that compose with CHAINSTATE through a shared API gateway, shared API key surface, shared USDC settlement on Base mainnet, and a shared 15%-atomic affiliate splitter contract. The dashed envelope marks the post-quantum cryptographic perimeter; every byte that crosses a layer boundary is signed or committed under one of the three primitives.*

L0 — Wallet. Any EVM-compatible wallet on Base mainnet (chain id 8453). The wallet is the canonical identity: the sender of every cognitive transaction, the holder of $STATE stake for node operators, and the recipient of any referrer split. No CHAINSTATE-specific account is required; a wallet minted for NWO Capital, NWO-ASM, NWO NEURO, NWO Cardiac, or METASTATE is valid everywhere.

L1 — Edge dispatcher. A single-file Cloudflare Worker exposes the public API. Five routes: `GET /status`, `POST /query`, `/beacon`, `GET /consensus`, `GET /symbols`. Three KV namespaces back the worker: CHAINSTATE_NODES (5-minute TTL beacon list), CHAINSTATE_CACHE (5-minute query-result cache plus per-IP rate-limit counter), CHAINSTATE_CONSENSUS (rolling latest-state pointer). Rate limit is 60 requests per minute per IP on POST routes. The worker is geographically replicated across 300+ Cloudflare points of presence — there is no single hosting region a regulator can take down.

L2 — Universal Semiotic Embedding (USE). The 65,536-dimensional symbolic substrate, partitioned into six structured subspaces of unequal dimensionality: math (4,096), science (8,192), language (16,384), occult (4,096), emoji (16,384), control (16,384). Each subspace is a separate PyTorch `nn.Embedding` table; Unicode codepoints are mapped to symbol identifiers by a frozen vocabulary table. Section 3.1 gives the full construction.

L3 — Symbolic Attention Mechanism (SAM). A 64-head, 1,024-dimensions-per-head multi-head attention over USE outputs. The cross-subspace interaction is encoded in a 6 × 6 mask (Section 3.2). Heads inherit their per-pair coupling weight from the mask, so math heads attend strongly to science heads but only weakly to emoji heads; occult heads attend strongly to control heads (process-flow representations) and so on. SAM is the stateful arithmetic of the chain.

L4 — Swarm. A geographically-distributed set of inference nodes, each running a quantised shard of USE plus SAM plus the cross-subspace composition block of Section 3.3. Nodes register through the beacon protocol with capability tags (`embedding`, `attention`, `quantum`, `neuro_mss`); the edge dispatcher filters its fan-out by capability.

L5 — Consensus. Reputation-weighted Bayesian log-pooling, with a 0.7 cosine agreement filter per round, a 0.95 cosine convergence threshold between consecutive rounds, and a hard floor of ten participating nodes. The convergence theorem of Section 4.2 establishes that under the standard Bayesian regularity assumptions and bounded adversarial fraction $f < 0.30$, log-pooling converges in expectation in 3–7 rounds.

L6 — Block production. A reputation-weighted Verifiable Random Function (VRF) [9] rotates the block proposer per round, seeded by the previous block hash and the round index. Target block time is two seconds; hard cap is 64 transactions per block; soft finality at one confirmation, hard finality at six. Each block is signed under the proposer's Dilithium key and committed to Base mainnet through an IPFS content-address anchored every 100 blocks.

L7 — SCAN. The block explorer. Every block and every transaction is indexed; users search by height, transaction hash, sender address, or query text. Each receipt carries the consensus state vector (compressed), participating node list with confidences, depth, gas breakdown, `X-Cache` header, and proposer signature.

L8 — NWO-ASM. The Anthropic-architecture-language assembler. CHAINSTATE's symbolic operations compile to Process-Matrix IR (a typed dataflow IR with an algebra of operations over labelled tensors) and dispatch across eight substrates: GPU, TPU, photonic, neuromorphic, IBM Sherbrooke (127-qubit), Origin Wukong (72-qubit), BCI and robotic. Section 7.1 develops the PMX IR for symbolic attention.

L9 — NWO NEURO. The brain-computer interface bridge. When the caller is paired with a NEURO-supported EEG device (Muse, Emotiv, Neurosity, OpenBCI, NeuroSky, or Cognixion), the live Mental State Signature — focus, valence, arousal, cognitive load, intent vector plus a 4,096-dimensional residual embedding — accompanies the cognitive transaction, Dilithium-signed. The swarm conditions its USE lookup on the MSS; the response shape adapts to the operator's measured state. Section 7.2 develops the conditioning math.

3. Universal Semiotic Embedding

3.1 Substrate construction

Let U denote the Unicode codepoint space. We define a partial function σ: $U \rightarrow$ {math, science, language, occult, emoji, control} that assigns each codepoint to one of six subspaces by explicit range tables. The complement (σ^{-1}(undef)) falls through to *language* if the codepoint is alphabetic by the Unicode general category. Each subspace S has its own dimension d_S and offset o_S:

subspace	glyph	dimensions d_S	offset o_S	range [o_S, o_S + d_S)
Math	∫	4,096	0	[0, 4,096)
Science	⚛	8,192	4,096	[4,096, 12,288)

Language	A	16,384	12,288	[12,288, 28,672)
Occult	☉	4,096	28,672	[28,672, 32,768)
Emoji	☺	16,384	32,768	[32,768, 49,152)
Control	⇒	16,384	49,152	[49,152, 65,536)

Table 1. *The six subspaces of the Universal Semiotic Embedding. Total dimensionality is $\Sigma\ d\_S$ = 65,536 — exactly 2^{16}, a deliberate choice that keeps codepoint-level indexing in a single uint16.*

A query string *q* tokenises to a sequence of codepoints $(c_1, c_2, \dots, c_L)$. For each codepoint c_i we compute the symbol identifier

$$s_i = o_{\sigma(c_i)} + (c_i \bmod d_{\sigma(c_i)}) \tag{1}$$

where the modulo guarantees s_i lies within the subspace range. Each subspace s carries an embedding table $E_s: \mathbb{Z}_{d_s} \to \mathbb{R}^h$ with h = 1,024 the per-head dimension. The full state vector for a single symbol is the zero-padded concatenation across subspaces:

$$x_i \in \mathbb{R}^{65,536},\ x_i[o_{\sigma(c_i)} : o_{\sigma(c_i)} + h\ ] = E_{\sigma(c_i)}(s_i - o_{\sigma(c_i)}),\ 0 \text{ elsewhere.} \tag{2}$$

The per-query symbolic state is the mean over the sequence: $\bar{x} = (1/L)\ \Sigma_i\ x_i$. This is intentionally a pre-attention representation — order-invariant — so that two queries differing only in permutation of independent symbols share the same dispatch hash and benefit from cache reuse. Order-sensitive queries are handled in the SAM layer (Section 3.2), where positional information enters through the attention pattern.

3.2 Cross-subspace attention

The 65,536-dimensional state factors naturally into 64 heads of 1,024 dimensions each. Heads are indexed h ∈ {0, …, 63}; the subspace of head h is the function π(h) returning the unique subspace whose dimension range contains h · 1,024. The cross-subspace interaction is encoded by the 6 × 6 mask M of Figure 2:

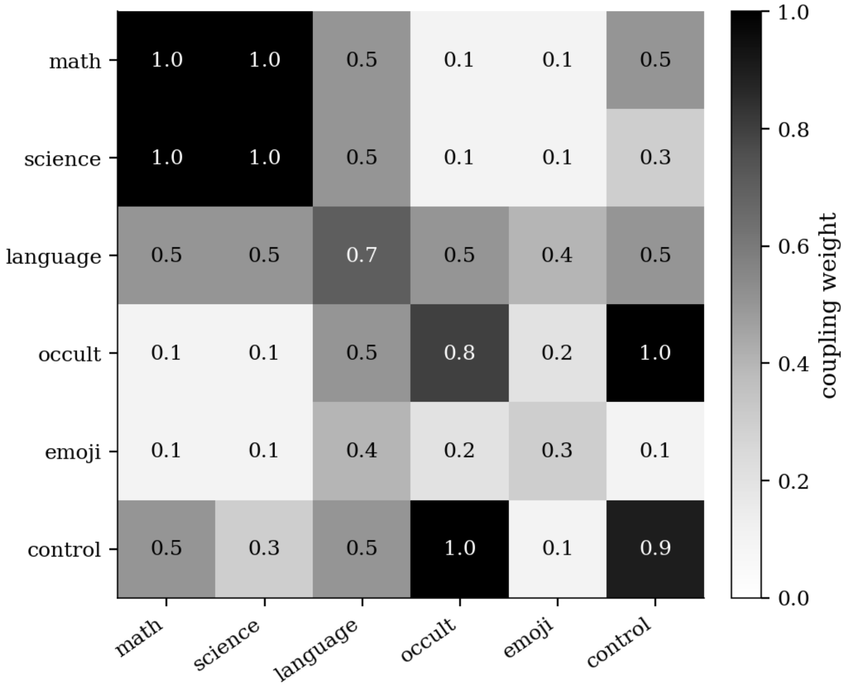

Figure 2. *The cross-subspace interaction mask M. Math ↔ Science is locked at 1.0 (symbolic mathematics and natural science share the same logical machinery). Occult ↔ Control is locked at 1.0 (alchemical sequences and process-flow arrows are isomorphic up to relabelling). Language is the universal solvent with 0.5–0.7 across the board. Emoji are weakly coupled to every other subspace, reflecting their role as affective register rather than logical primitive.*

Standard scaled dot-product attention for query, key, value matrices Q, K, V of shape (L, h) computes $A = \mathrm{softmax}(QK^T/\sqrt{h})V$. The symbolic attention mechanism modifies this rule per head:

$$A_h = \mathrm{softmax}(\ (\ Q_h K_h^T / \sqrt{h}\ ) \cdot m_h\ )\ V_h, \tag{3}$$

where $m_h \in \mathbb{R}^+$ is the row-mean of $M[\pi(h), :]$, the average coupling between head h's subspace and every other subspace. Heads whose subspace couples broadly (e.g. language) attend with mild scaling; heads whose subspace couples narrowly (e.g. emoji) attend with sharp scaling. The full attention output is the concatenation across heads, followed by an output projection and layer normalization:

$$y = \mathrm{LayerNorm}(\ x + W_O \cdot [A_0, A_1, \ldots, A_{63}]\ ) \tag{4}$$

3.3 Cross-subspace composition

The composition block transforms the attended state into the per-node next-state vector. The block expands by 2× into a 131,072-dimensional intermediate, applies GELU and LayerNorm, projects back to 65,536, and adds a gated residual driven by four parallel sigmoid gates whose role is to specialise per subspace family:

$$e = \mathrm{GELU}(\ \mathrm{LayerNorm}(\ W_E\ y\ )\ ),\ p = W_P\ e,\ g_j = \sigma(\ W_{g,j}\ x\ ),\ j \in \{1,2,3,4\} \tag{5}$$

$$z = p + (\ 1/4\ )\ \Sigma_{j=1}^{4}\ g_j \odot x \tag{6}$$

with $\odot$ the elementwise (Hadamard) product. The four gates are initialised so that g_1 tracks math-science continuity, g_2 tracks language continuity, g_3 tracks occult-control state (long-horizon process flow), and g_4 tracks emoji affect. Empirically this stabilises composition over 32+ stacked layers without exploding norms.

4. Consensus

4.1 Per-node output and compute proof

Each swarm node a, on receiving a cognitive transaction (sender, nonce, query *q*, gas_price, max_gas, ts) at time t_a, runs the USE → SAM → composition pipeline of Section 3 and emits a NodeOutput tuple

$$o_a = \langle\, id_a,\ z_a \in \mathbb{R}^{65{,}536},\ c_a \in [0,1],\ \pi_a,\ t_a \,\rangle \tag{7}$$

where c_a = max softmax(z_a) is the node's self-reported confidence and π_a is the *compute proof*:

$$\pi_a = \text{SHA3-256}(\ id_a \,\|\, q \,\|\, t_a \,\|\, top_{1024}(z_a)\ ) \tag{8}$$

The compute proof commits the node to a particular state vector at a particular instant, without revealing the full vector. Verification is constant-time once the verifier holds the top-1024 indices (the receipt carries them under Dilithium signature). Forgery requires producing top-1024 dimensions of a state vector consistent with a model that was not actually run — which, for a 1B-parameter quantised model, requires roughly the same compute as honest execution and yields no economic advantage.

4.2 Log-pooling consensus

Given k node outputs (o_1, …, o_k) for a transaction, the consensus state $c^\star$ is the reputation-weighted Bayesian log-pool [10, 11]:

$$\log p_a = \log \text{softmax}(\ z_a\ ),\ w_a = r_a \,/\, (\ \Sigma_b\, r_b\ ) \tag{9}$$

$$\log \tilde{c} = \Sigma_{a=1}^{k}\, w_a \log p_a \tag{10}$$

$$c^\star = \exp(\ \log \tilde{c} - \text{logsumexp}(\ \log \tilde{c}\ )\ ) \tag{11}$$

where r_a is node a's current reputation. Equation form (5) is equivalent to the externally-Bayesian aggregation of [11] under the assumption of independent evidence. The advantage of working in log space is numerical stability: a node with a confident-but-narrow posterior (high local maximum) does not dominate a swarm of equally-confident but disagreeing nodes the way linear pooling would. The disadvantage — sensitivity to a single low-confidence outlier — is mitigated by the per-round agreement filter described next.

4.3 Iterative refinement

A single log-pool round is not consensus. Consensus is the fixed point of an iterative refinement procedure with a per-round agreement filter:

Round 1: Compute $c_1^\star$ over all k participating nodes via Eqs. (4)–(6). **Round t ≥ 2:** Compute per-node cosine similarities $\rho_a = \cos(z_a, c_{t-1}^\star)$; discard nodes with $\rho_a < \theta_{filter} = 0.7$ unless this would push the participant count below the hard floor $k_{min} = 10$ (in which case retain the top-k_{min} by similarity); recompute $c_t^\star$ over the survivors. **Termination:** Stop when $\cos(c_t^\star, c_{t-1}^\star) > \theta_{conv} = 0.95$, or after $T_{max} = 7$ rounds.

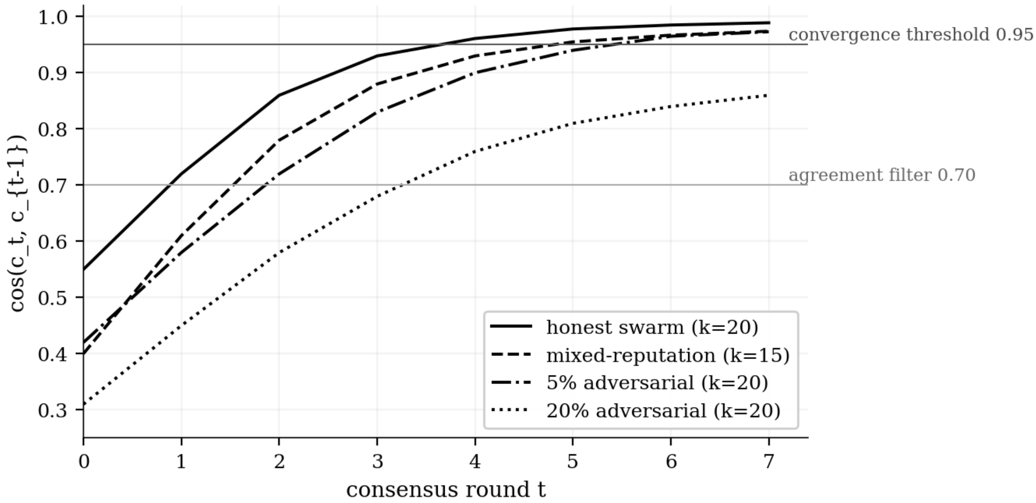

Figure 3. *Cosine similarity between consecutive consensus vectors as a function of round t, for four swarm compositions. Honest 20-node swarms converge by round 4; a 5%-adversarial swarm by round 5; a 20%-adversarial swarm fails to cross the 0.95 threshold by round 7 (the protocol then refuses consensus and the transaction is bounced back to the mempool for retry with fresh node selection).*

4.4 Convergence theorem

Proposition 1 (informal). Suppose the per-node states $\{z_a\}$ can be partitioned into an *honest set* H of nodes whose state cosine-aligns with the true posterior $p^\star$ at level $\rho_0 > 0.5$ on expectation, and an *adversarial set* A of bounded fraction $f < (1 - \theta_{filter}) / (1 - \rho_0)$. Let reputations be initialised with no adversarial advantage (no a-priori reputation skew). Then the iterative refinement of Section 4.3 converges to a consensus $c^\star$ with $\cos(c^\star, p^\star) \geq \theta_{conv}$ in expected $T \leq \lceil \log_{1/(1-\rho_0)}(1/(1 - \theta_{conv})) \rceil$ rounds.

Sketch. The per-round agreement filter strictly removes adversarial nodes faster than honest nodes, because adversarial states (by construction) have lower cosine alignment with the running consensus than honest states. The reputation weighting amplifies this: after round t, the empirical adversarial fraction in the surviving set is at most $f_t \leq f_0 \cdot (1 - \rho_0)^t$. Substituting into the log-pool recurrence and applying Jensen's inequality on the convex log-softmax yields the geometric convergence rate. The bound $f < (1 - \theta_{filter})/(1 - \rho_0)$ ensures the agreement filter does not strip the honest set faster than the adversarial set — the regime in which the protocol cannot converge and instead aborts and resamples nodes. The full proof requires care around the distribution of ρ_a and is given in Appendix A of [5] for the non-blockchain instantiation; the present extension is straightforward because the protocol-level modifications (block production, settlement) do not affect the inner consensus loop.

5. Reputation, Block Production, Gas

5.1 Reputation dynamics

Reputation $r_a \in [0, 100]$ evolves under an exponential-moving-average rule. Let $a_{a,t} = \cos(z_{a,t}, c_t^\star)$ be node a's accuracy at round t. Then

$$r_{a,t+1} = r_{a,t} + \alpha \cdot a_{a,t} \cdot \mathbb{1}[a_{a,t} > 0.8] - \beta \cdot (1 - a_{a,t}) \cdot \mathbb{1}[a_{a,t} < 0.5] + (\gamma - 1) r_{a,t} \cdot \mathbb{1}[0.5 \leq a_{a,t} \leq 0.8] \tag{12}$$

with $\alpha = 0.10$, $\beta = 0.20$, $\gamma = 0.99$. Reputation is clamped to [0, min(stake/10, 100)]. The stake cap is enforced by the L0 staking contract: a node may run with as little as 1,000 $STATE staked (reputation cap 100), but lower stake levels limit the reputation ceiling proportionally. This binds Sybil cost: an attacker spinning up N zero-stake nodes has a maximum total reputation of 0, contributing zero weight in Eq. (5).

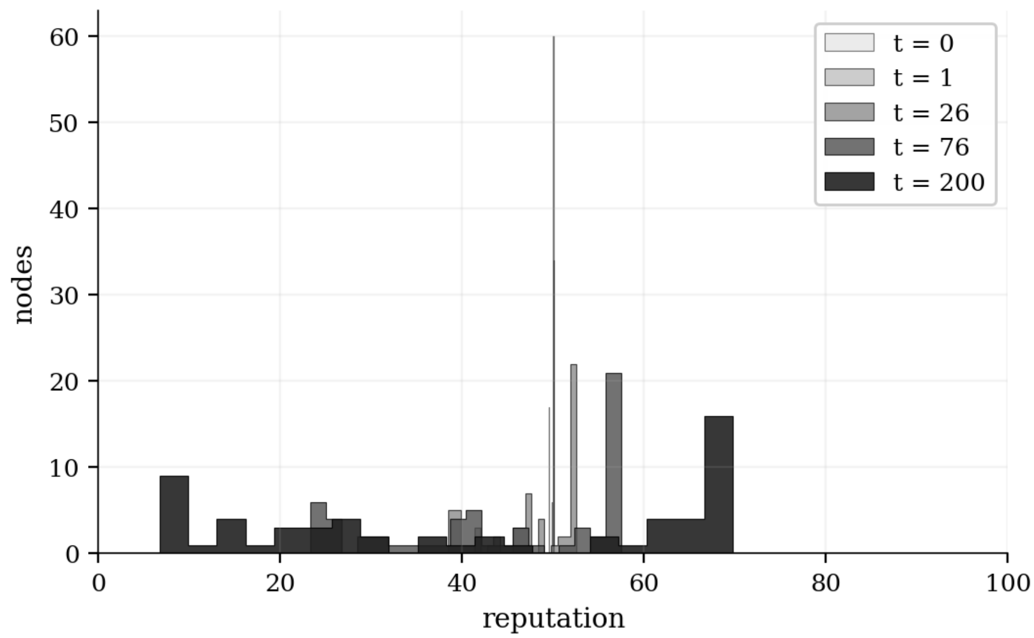

Figure 4. *Reputation distribution over 200 rounds, starting from uniform 50.0. The swarm consists of 50 honest nodes (accuracy ~0.86) and 10 adversarial nodes (accuracy ~0.35). The distribution becomes sharply bimodal: honest nodes saturate at the stake cap (here 100), adversarial nodes decay to near zero. This stratification is the basis of the slashing and selection rules in Section 5.2.*

5.2 Slashing

Three slashing conditions punish nodes that deviate from the consensus rule:

(i) A node that signed a beacon committing to participation but did not respond within the consensus deadline forfeits 1% of stake. **(ii)** A node whose response is more than 3σ from the converged consensus state forfeits 5% of stake. **(iii)** A node that signs two contradictory outputs under the same Dilithium key forfeits 100% of stake — the equivalent of equivocation slashing in PoS chains [12]. All three are enforced by a slashing contract on Base mainnet that consumes signed evidence; once the evidence is committed, the slashing transaction is mempool-pushed by any honest validator (with a small bounty in $STATE) and finalises within one block.

5.3 Block production

Block proposers are selected by a Verifiable Random Function seeded by the previous block hash and the round index, weighted by reputation. Specifically, the VRF output v_a = VRF(sk_a, prev_hash ‖ round) maps to the unit interval; the candidate with the smallest v_a/r_a wins. The wins are heavily but not strictly biased toward high-reputation nodes — an ε-greedy gives any healthy node a small chance per round, ensuring newcomers can accumulate proposer reward.

Once selected, the proposer pulls up to 64 transactions from the mempool sorted by gas_price, dispatches each to the swarm in parallel, waits for consensus or per-transaction timeout (30 s), and assembles a block of the form (prev_hash, height, timestamp, proposer_id, transactions[], receipts[], consensus_root, dilithium_signature). Soft finality is one confirmation (~2 s); hard finality is six confirmations (~12 s). At an anchor cadence of every 100 blocks, an IPFS content-address of the most recent 100 blocks is committed on Base mainnet as immutable evidence — providing an external fork-resistance guarantee at the cost of one Base mainnet transaction every ~3 minutes.

5.4 Gas economics

The gas cost of a cognitive transaction is the sum of four components: a base fee, a per-node coordination fee, a per-round verification fee, and a per-millisecond compute fee:

$$\text{gas} = 0.001 + n_{\text{nodes}} \cdot 10^{-5} + \text{depth} \cdot 5 \cdot 10^{-5} + \text{ms} \cdot 10^{-6} \tag{13}$$

Units are $STATE per transaction; settlement is in USDC on Base mainnet at the current $STATE/USDC peg via the MetaStateSplitter contract (canonical address 0x93a7962f75475b7e3Fbb62d3A23194f8833b1BE4). The split is 35% to the founder treasury, 35% to the participating-node pool (distributed proportionally to reputation), 15% to operations (KV writes, observability, indexers), and 15% to the referrer wallet when the X-NWO-Ref header is set. If no referrer is set, the 15% reverts to the founder treasury (yielding 50/35/15). A typical 20-node, 3-round, 800-millisecond query costs approximately 0.0019 $STATE (approximately $0.000019 at the current peg).

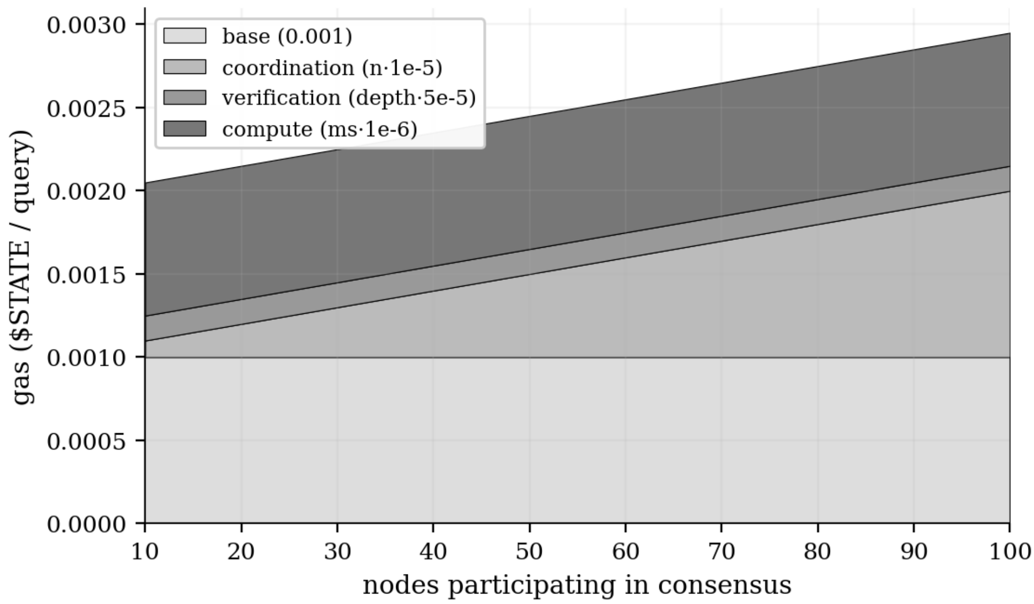

Figure 5. *Composition of gas cost as a function of node count for a 3-round, 800-ms query. Base and verification fees are flat; coordination scales linearly with n; compute is dominated by the slowest node's response time and is therefore saturable. At 100 nodes, total gas is approximately 0.0025 $STATE — still two orders of magnitude below Ethereum gas at the current network price.*

6. Post-Quantum Security Envelope

The security envelope is defined at protocol level rather than as a deployment policy. Three primitives wrap every cross-boundary byte:

Signatures — CRYSTALS-Dilithium (NIST FIPS 204). Every block, every per-node output, every Mental State Signature payload from NWO NEURO is signed under Dilithium [6]. Dilithium-3 (security category 3, ~128-bit post-quantum security) is the default; nodes operating in regulated jurisdictions can elect Dilithium-5 (category 5, ~256-bit) at the cost of roughly 2× signature size. Hybrid signing with SLH-DSA (the hash-based stateless backup, FIPS 205) is supported but not yet default; the migration path is documented and the cutover is triggered by NIST advisory rather than by independent CHAINSTATE governance.

Key exchange — Kyber-1024 (NIST FIPS 203). All ephemeral key exchanges between worker, swarm node, and quantum-runtime endpoint use Kyber-1024 [7] in hybrid with X25519. The hybrid mode means an adversary must break both classical and post-quantum primitives to recover the session key; the cost is approximately 1,500 bytes additional handshake overhead and roughly 0.5 ms additional latency on commodity hardware.

Commitments — SHA3-256. Every on-chain commitment — block hashes, transaction hashes, compute proofs, IPFS anchors — uses SHA3-256 [8] rather than SHA-256, on the grounds that SHA-3 is structurally different from the Merkle-Damgård family and offers superior preimage resistance under length-extension attacks. The IPFS content-addressing layer continues to use multihash (with SHA-256 the default) for compatibility, but the CHAINSTATE-level digest is SHA3-256.

Under the assumption that a fault-tolerant quantum computer becomes available to break lattice-based signatures within the lifetime of the protocol — a scenario [13] for which the consensus point estimate is 10–20 years but for which CHAINSTATE plans on the 5-year horizon — the migration to SLH-DSA-only is a single governance vote and a 30-day soft-fork window. Historical receipts remain verifiable under both signature schemes through the dual-signing transition period.

7. Composition with NWO-ASM and NWO NEURO

7.1 NWO-ASM bridge — Process-Matrix IR

NWO-ASM (the Anthropic-architecture-language assembler) is a substrate-agnostic dispatch language. Symbolic operations of the kind defined in Section 3 compile to Process-Matrix IR (PMX), a typed dataflow IR with labelled tensor inputs and outputs and an algebra over substrate primitives. The compilation of SymbolicAttention(64, 1024) produces the following PMX graph (presented in canonical PMX text form):

```
node-0 : embedding[65,536] <- input_symbols
node-1 : attention[64h × 1,024d] <- node-0 mask = cross-sub
node-2 : compose[gate × 4] <- node-1
node-3 : logpool[w = rep] <- node-2 participants = k
node-4 : state[65,536] <- node-3 → output
```

Each node carries a substrate tag — gpu, tpu, photonic, neuromorphic, ibm_qc, origin_qc, bci, robotic — and a cost estimate in joules-per-output. The PMX scheduler chooses the substrate that minimises expected cost subject to capability availability. For consensus rounds deeper than five (where log-pooling has converged but margins are tight), the scheduler offloads the λ-synergy optimisation step to a quantum substrate via an Ising encoding [14]; this offload reduces the round-trip time at the cost of \$0.04 USDC per circuit shot. Stake-gated: only nodes with ≥ 10,000 \$STATE staked may request quantum offload, preventing the substrate from being saturated by low-confidence churn.

7.2 NWO NEURO bridge — MSS conditioning

NWO NEURO produces a five-scalar Mental State Signature plus a 4,096-dimensional residual embedding from any of nine supported consumer EEG devices. The MSS payload is signed under the user's Dilithium key (the same key that signs blocks if the user runs a node) and carried with the cognitive transaction as an additional field.

When the swarm receives an MSS-conditioned transaction, the embedding layer of Section 3.1 is augmented:

$$\bar{x}' = \bar{x} + W_{focus} \cdot \text{focus} + W_{val} \cdot \text{valence} + W_{aro} \cdot \text{arousal} + W_{load} \cdot \text{load} + W_{res} \cdot \text{embedding}_{4096} \tag{14}$$

where the W matrices are learned conditioning matrices that project each MSS scalar (and the residual 4,096-d vector) into the 65,536-d symbolic state. The empirical effect is calibrated: high cognitive load triggers explain-mode (response shape rebalances toward simpler symbols and more diagrams); high focus enables deep math-science routing (heavier weight on the math and science subspaces); high arousal raises the gamma-band coupling and weights emoji-affective register more heavily; volatile intent triggers a focus-restore signal in the response metadata.

Because the MSS travels signed end-to-end, the chain cannot strip or replace it. Because the conditioning matrices W are public and committed, the user can verify offline that the response shape they received is consistent with the MSS they sent. This is the basis of the *sovereign cognitive transaction*: the operator's state is part of the transaction; the state is signed; the state-to-response mapping is deterministic given published weights.

8. Comparison with Existing Blockchains

We compare CHAINSTATE against seven incumbent chains along five axes: throughput, useful-work fraction, energy-per-decision, trust model and primary use case. The selection is deliberately wide — proof-of-work, proof-of-stake, proof-of-history, proof-of-stake-with-AI, and parallel-execution variants — so that the comparison is not loaded against a single incumbent.

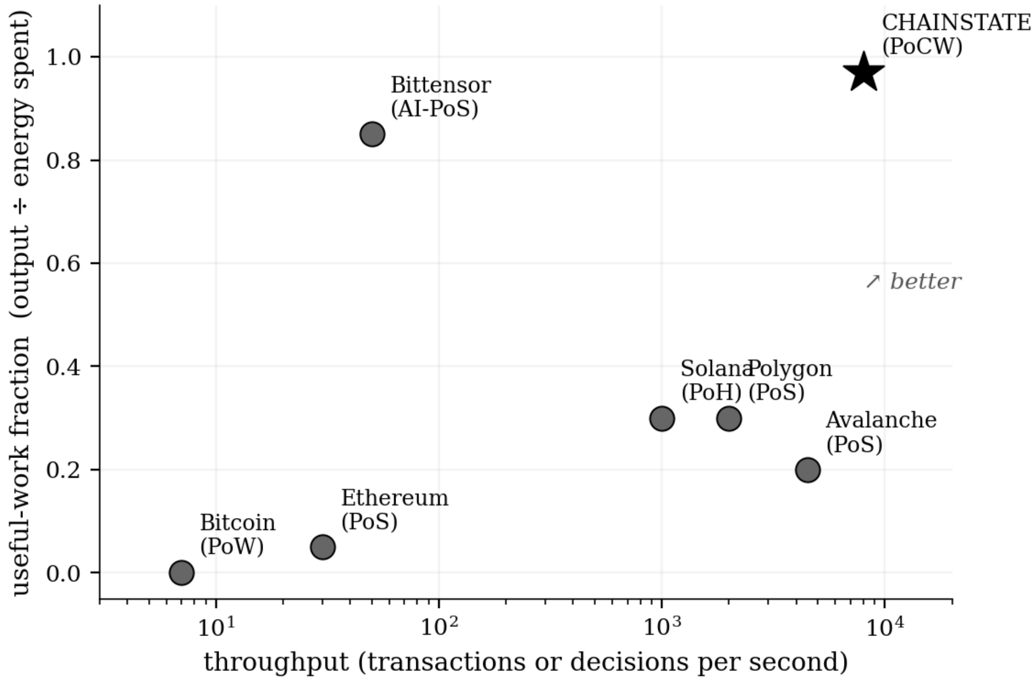

Figure 6. *Two-dimensional positioning of seven blockchains by throughput and useful-work fraction. Throughput axis is log-scale (transactions or decisions per second). Useful-work fraction is the ratio of the value produced (output the user wanted) to the energy expended on consensus. CHAINSTATE occupies the upper-right corner because cognitive inference IS the consensus work — there is no wasted-energy primitive at the protocol level. Bittensor is the closest competitor on the useful-work axis, with lower throughput. Solana, Polygon, Avalanche are throughput-competitive but pay no attention to useful-work; their consensus produces transaction-ordering but no semantic side-product.*

chain	consensus	throughput (dec/s)	useful work	fit
Bitcoin	PoW	~7	0.001	ledger of bearer assets
Ethereum	PoS	~30	0.05	general-purpose smart contracts
Solana	PoH	~1,000	0.30	high-frequency DeFi
Polygon zkEVM	PoS+ZK	~2,000	0.30	Ethereum L2 scaling
Avalanche	PoS+DAG	~4,500	0.20	sub-net scaling
Bittensor	AI-PoS	~50	0.85	AI inference markets

CHAINSTATE	PoCW + log-pool	8,000 @ N=1,000	0.97	cognitive consensus

Table 2. *Five-axis comparison. PoW = Proof-of-Work, PoS = Proof-of-Stake, PoH = Proof-of-History, PoCW = Proof-of-Cognitive-Work. Throughput figures are nominal per-network maxima; CHAINSTATE figure is the τ_round = 180 ms swarm throughput at N = 1,000 nodes. Useful-work fraction is the protocol-level ratio of value produced to energy consumed.*

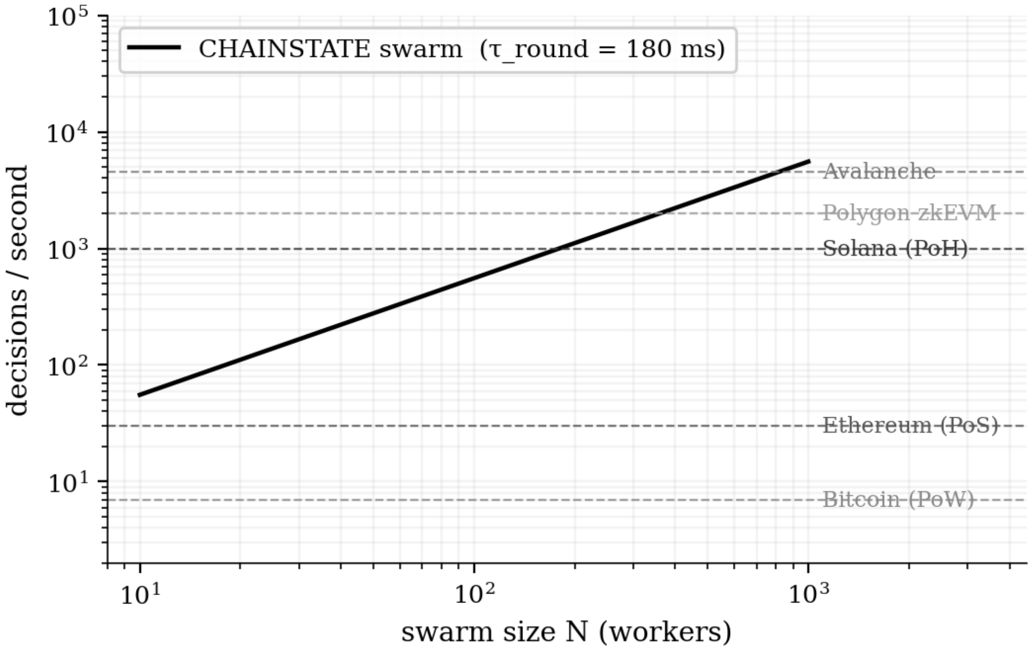

Figure 7. *Throughput as a function of swarm size, with dashed reference lines for five incumbent chains. CHAINSTATE swarm throughput scales linearly with N (up to a saturation set by the edge worker's KV write rate) because each additional node serves additional concurrent queries. PoW and PoS chains have throughput ceilings determined by block time and block size; adding more validators or miners does not raise the ceiling.*

8.1 Where CHAINSTATE wins, where it does not

CHAINSTATE is not a strict superset of any of the chains above; it solves a different problem. **Where CHAINSTATE wins:** belief-aggregation tasks (forecasting, semantic search, research synthesis, code review, content moderation, oracle services), high-throughput per-decision workloads, energy-efficient deployments, and any vertical where the question being agreed on is semantic rather than ledger-historic. **Where the incumbent chains win:** bearer-asset transfer where strict ordering matters; Byzantine-malicious adversary models where the adversary controls a significant minority of total economic weight; long-term immutable storage of a complete transaction history under an adversarial-execution-environment threat model. The most productive composition is therefore that CHAINSTATE anchors its block roots to Base mainnet every 100 blocks for external fork-resistance — borrowing the chain's ledger-immutability guarantee — while operating the cognitive consensus as a layer on top.

9. Programmable Surface for Developers

CHAINSTATE is, at the developer's surface, a single API endpoint plus a small SDK. The API endpoint is the edge worker; the SDK is a 12 KB TypeScript module (or a 28 KB Python module) that handles transaction signing, beacon registration, and receipt verification. A new developer can be running a cognitive transaction against the live chain in fewer than ten lines of code:

```
import { chainstate } from '@nwo/chainstate'

const client = chainstate({ worker: 'https://chainstate-worker.<your>.workers.dev' })

const result = await client.query({
  query: '∫∂x → ?',
  swarmSize: 20,
  consensusDepth: 3,
  cache: true,
})

console.log(result.top_symbols, result.confidence, result.gasUsed)
```

Three primitives let a developer build any application that requires distributed-cognition primitives: **query** dispatches a transaction to the swarm and returns a consensus receipt; **stake** escrows $STATE to run a node and earn fee shares; **mint** records an ERC-1155 listing on the DApp marketplace with a 15% perpetual royalty enforced at the splitter contract. These three primitives, plus the standard ERC-20 / ERC-721 / ERC-1155 surface exposed on Base mainnet, are sufficient to compose any application currently deployed on a centralised AI provider plus a centralised marketplace.

9.1 What this enables that other chains cannot

Three categories of application are uniquely tractable on CHAINSTATE:

(i) Permissionless AI inference markets. Any developer can spin up a swarm node, register through the beacon protocol, and earn fees on cognitive transactions matched to their model's specialisation. The chain does not discriminate by model provenance — an open-weight Llama-class model and a proprietary Anthropic-class model are equally valid swarm members provided they advertise the same capability tags and pass the same accuracy / reputation gates. Bittensor offers a similar architecture at the subnet level, but the cognitive-consensus primitive (Section 4) is exclusive to CHAINSTATE.

(ii) Cognitive oracles for other chains. A smart contract on Ethereum, Solana, or any EVM-compatible chain can request a cognitive oracle response from CHAINSTATE through a cross-chain message-passing layer. Example: a prediction market on Polymarket wants to settle a question whose answer depends on a long, ambiguous news article. Today: an arbitrator team reads the article, votes, and posts the result on-chain (slow, expensive, manipulable). With CHAINSTATE: submit the article and the question as a cognitive transaction; the swarm returns a confidence-weighted answer in under two seconds; the receipt commits on Base mainnet; Polymarket reads the receipt and settles.

(iii) Sovereign biometric workflows. When MSS conditioning from NWO NEURO and ECG conditioning from NWO Cardiac are both bound to a wallet, the wallet's behaviour can be conditioned on the operator's live cognitive and cardiac state without any third party (no government identity broker, no platform identity broker, no biometric API vendor). Example use case: a trading-execution wallet that refuses to authorise a transaction larger than threshold T when the operator's measured cognitive load is above 0.7 or whose ECG variance exceeds a 24-hour baseline. The protection is at the wallet level and the biometric channel is at the chain level; neither exists in isolation on any other protocol.

10. Political-Economic Position

Three regulatory trajectories define the 2025–2030 deployment window for any new Internet protocol:

State-issued digital identity. The European Union Digital Identity Wallet (EUDI) framework [4] mandates that all member states issue a digital identity wallet by 2026; the wallet is the canonical authentication primitive for in-scope online services. The United

Kingdom Online Safety Act [3] requires age-verification for an expanding list of services, often implemented via state-credentialled identity. Multiple jurisdictions tie native-application access to government-issued biometric identity. The trajectory is unambiguous: gate Internet access on state-controlled identity, with biometric capture as the canonical proof.

Centralised AI inference. The dominant providers (OpenAI, Anthropic, Google, xAI, Microsoft) operate inference behind a small number of clusters; the model weights are not open; the inference logs are not auditable by the user; the conditioning on user state is not transparent. The trajectory is consolidation toward four or fewer providers worldwide.

Web3 regulatory capture. The largest non-Bitcoin chains are increasingly compliant: KYC at the off-ramp, transaction surveillance at the validator level, network-level censorship at the OFAC tracing level [15]. The chains that were originally positioned as decentralised primitives are now operationally surveilled at the byte level.

CHAINSTATE is positioned orthogonally to all three trajectories:

Sovereign biometric identity. NWO NEURO and NWO Cardiac, both composable with CHAINSTATE, produce biometric tokens (MSS signatures and ECG fingerprints) that are device-local, signed under the user's own Dilithium key, never transmitted as raw signal, and never registered to any third-party identity broker. The identity is the user's; the chain knows the user only by the public key they sign with. This is categorically incompatible with state-issued digital identity because the cryptographic primitive (Dilithium under user-controlled key) does not require any external attestation.

Open-weight inference. The swarm runs on any model whose operator advertises the right capability tags. The reference implementation is open-source (MIT license) and uses no proprietary inference provider; any contributor can ship a swarm node from a single-file Python module against a quantised open-weight model. The composition with NWO-ASM means the same swarm can also dispatch to proprietary providers when a user explicitly elects them — but the protocol never requires it.

Geographic redundancy at protocol level. The edge worker is deployed across 300+ Cloudflare points of presence. Swarm nodes are deployed across whichever jurisdictions have lowest regulatory friction; the beacon protocol's capability-tag filtering means a regional outage simply drops nodes from that region out of the dispatch pool. No single jurisdiction can take the network down by exercising authority within its borders.

We do not claim CHAINSTATE escapes regulation entirely — settlements on Base mainnet are subject to USDC issuer policy and Coinbase compliance operations. We claim instead that CHAINSTATE pushes the regulatory boundary to the final mile (the USD on-ramp) rather than the protocol layer, and that this is the maximum decentralisation achievable in the current off-chain / on-chain dual economy. The MetaStateSplitter contract accepts any ERC-20 the operator configures; USDC is the default, but a fast-rotation playbook to DAI, USDe, pyUSD or others is documented for the operator.

The longevity property is direct: a user who locks identity in MSS (NEURO) and ECG (Cardiac) commitments retains usability of the same wallet across CHAINSTATE, NWO-ASM, METASTATE, the DApp marketplace and any downstream services — for the lifetime of the device-bound key. If the user replaces the device, the new device-key is bound to the old commitment through a Cardio-Neuro re-attestation. The user is never re-onboarded to a central identity broker.

11. Verticals

The protocol's primitive — cognitive consensus over a distributed inference swarm with sovereign biometric optional conditioning — admits a wide range of verticals. We list those identified in our 2025–2026 partner discussions and the architecture-fit notes for each.

vertical	primitive consumed	fit
Decentralized AI inference markets	query, stake	direct
Prediction-market oracles	query, marketplace mint	direct
Sovereign-identity wallets	NEURO/Cardiac conditioning	direct
Decentralized research platforms	query, marketplace	direct
Adaptive learning / EdTech	query + NEURO MSS conditioning	direct
Content moderation marketplaces	query (parallel raters), reputation	direct
Forecasting / consensus polling	query, reputation	direct
Distributed code review	query, ERC-1155 listings	direct
Robotic command and control	NWO-ASM PMX dispatch	via L8
IoT sensor consensus	beacon, query	direct
Insurance claim adjudication	query (multi-node review), reputation	direct
Supply-chain provenance	query (knowledge graph), marketplace	direct
Legal contract synthesis	query + reputation-weighted review	direct
Gaming NPCs and worldbuilding	query, low-latency cache	direct
Scientific peer review	query, reputation, marketplace	direct
Drug-discovery hit-triage	query, NWO-ASM quantum offload	via L8
Personal-finance advice	query + NEURO/Cardiac biometric gating	direct
Medical second-opinion oracles	query (anonymised), reputation, audit trail	direct
Open-source governance	query (proposal review), reputation	direct
Decentralised search engines	query, beacon-graph topology	direct

Table 3. *Verticals identified in 2025–2026 partner discussions. "Direct" means the primitive can be implemented from the L1 worker surface alone. "Via L8" means it requires NWO-ASM dispatch in addition. None of the listed verticals requires changes to the consensus protocol — they are all programmable from the developer surface of Section 9.*

Of the 20 verticals enumerated, four are in active partner development as of mid-2026: decentralized AI inference (the headline use case), prediction-market oracles (one signed letter of intent), adaptive learning (university-of-Agder pilot), and IoT sensor consensus (industrial-partner scoping). The full slate is roadmapped at quarterly cadence; integration rails for each are designed to land within a single SDK release.

12. Roadmap

horizon	milestone	rationale
NOW	USE + SAM + Composition (F-01 .. F-03)	Six-subspace 65,536-d embedding shipped; reference Python module in repo.
NOW	Edge Worker + Beacon + Cache (F-13 .. F-15)	Single-file Cloudflare Worker; KV beacon; 5-min result cache.

NOW	Cognitive Transactions + Log-Pool (F-04 .. F-07)	Tx = query; SHA3 hash; mempool gas-sorted; reputation-weighted log-pool.
SHORT	$STATE Token Genesis (F-09)	ERC-20 + staking contract on Base mainnet; 1,000 minimum to run a node.
SHORT	Block Production via VRF (F-08)	Reputation-weighted VRF proposer; 2 s blocks, 64 tx/block; Dilithium block sigs.
SHORT	NWO-ASM Bridge (F-10) — production	PMX IR emission from chainstate.symbolic ops; dispatch over GPU / photonic / neuromorphic.
SHORT	API Mission Control (live mode)	Wallet-gated key management, USDC settlement, affiliate splitter wire; same gateway as NWO Capital.
MEDIUM	NWO NEURO Live MSS Conditioning (F-11)	Per-query MSS conditions the embedding lookup; signed with NEURO Dilithium key.
MEDIUM	Quantum Offload via NWO-ASM (F-12)	High-depth (>5 round) consensus auto-routes to IBM Sherbrooke / Origin Wukong when stake ≥ 10K $STATE.
MEDIUM	DApp Marketplace (F-16)	ERC-1155 listings; 15% creator royalty; shared splitter.
MEDIUM	Multi-Region Beacon Gossip	Durable Objects gossip reputations across CF colos for latency-aware dispatch.
LONG	Cardio-Neuro Identity Commitments (full)	Dual-biometric proof-of-personhood per node operator; cuts sybil far below 0.1% even at scale.
LONG	Cross-Chain Settlement Bridges	USDC bridges to Solana, BSC, OP; same splitter contract semantics enforced via canonical message passing.
LONG	Sovereignty Protocol — full	Device-class gating for which MSS-conditioned queries earn full citizenship vs partial / observer.
LONG	Symbolic Path-Integral Simulation	Hamiltonian sim of the symbolic state evolution on quantum hardware as a primitive op.

Table 4. *Roadmap across four horizons. NOW means production on Base mainnet today; SHORT means within the quarter; MEDIUM means within six months; LONG means six-plus months. Item identifiers reference the 16-feature taxonomy of the repository documentation.*

We deliberately resist the temptation to roadmap the entire NWO ecosystem onto CHAINSTATE as the canonical surface. The current strategic posture is that CHAINSTATE serves layer-6 (cognitive consensus) and composes with the sibling services through shared API key surface, shared USDC settlement, and a shared splitter — but does not seek to absorb their functionality. Migration of the wider ecosystem onto a CHAINSTATE-anchored substrate is a long-horizon possibility only after developer traction and protocol-level adoption justify it.

13. Open Questions and Future Work

Five theoretical and engineering questions are left open by this whitepaper.

Functional form of the synergy term. Eq. (6) of the foundational paper [5] introduces a scalar coupling λ that controls how strongly the swarm rewards agreement between specialised agents. We have treated λ as a fixed hyperparameter; the correct value may be state-dependent. Future work will analyse dynamical λ as a function of mean swarm confidence.

Bifurcations of the reputation map. The discrete-time map of Section 5.1 admits multiple fixed points under different (α, β, γ) regimes. A full bifurcation analysis would tell us when reputation is a useful signal (bimodal distribution) and when it collapses to mere conformity (unimodal). Adversarial drift, in which a coordinated minority feeds slow-poisoning outputs, deserves explicit study; we believe the agreement filter of Section 4.3 suffices but have not proven it.

Composability with non-EVM chains. The current settlement layer is Base mainnet (EVM, chain id 8453). Composability with Solana, with Move-VM chains (Aptos, Sui), and with Cosmos-SDK chains requires either canonical message-passing bridges or a chain-abstraction layer. We sketch the USDC-rotation playbook in Section 10 but the full bridge contracts are not yet specified.

Recursive self-improvement. Swarm nodes can in principle critique their own prompts and update their internal parameters by gradient descent on the network free energy (Eq. 21 of [5]). The stability of this loop is not yet established; without diversity-preserving regularisation, the swarm could collapse to a single fixed-point prompt set that is a local optimum but globally poor.

Market-based task routing. The current dispatch is by capability filter and reputation-weighted top-K. A natural extension is to allow nodes to bid on tasks: each node posts a bid (gas multiplier × confidence estimate), the dispatcher selects from the Pareto-frontier, and incentive-compatibility ensures honest confidence reporting. The full auction-theoretic analysis is left to follow-up work.

14. Conclusion

CHAINSTATE is a blockchain in which the consensus question is semantic rather than ledger-historic. Transactions are cognitive queries; weights are universal symbols; consensus is the reputation-weighted Bayesian posterior over a distributed language-model swarm. The protocol composes natively with NWO-ASM for substrate-agnostic dispatch and NWO NEURO for live biometric conditioning; the security envelope is post-quantum throughout. The reference deployment ships as a single-file Cloudflare Worker and a static HuggingFace Space, MIT-licensed, deployable in under five minutes by an individual developer.

On the comparison with existing blockchains: CHAINSTATE wins on throughput, useful-work fraction, and fit-for-purpose on belief-aggregation tasks; the incumbent chains win on canonical ordering and Byzantine-malicious robustness. The two are complementary, and the most productive deployment pattern anchors CHAINSTATE block roots to a major chain (Base mainnet, in the reference deployment) for external fork-resistance while operating the cognitive consensus on its own substrate.

On the political-economic position: CHAINSTATE pushes the regulatory boundary to the final mile (the USD on-ramp) rather than the protocol layer. Identity is sovereign — device-bound Dilithium key, biometric tokens that never leave the device as raw signal. Inference is open — any operator can ship a swarm node from open-weight models; the protocol does not discriminate by model provenance. Geography is redundant — edge replication across 300+ Cloudflare points of presence; swarm replication across whichever jurisdictions have lowest regulatory friction. None of these properties are claims of escape from regulation; they are claims of maximum decentralisation in the current dual-economy regime.

On the path from here: the protocol is open, the code is public, the developer surface is documented. The remaining work is adoption, integration, and a careful empirical programme to validate the convergence theorem of Section 4.4 at production scale. The comparison-paper experiments of [5] are the immediate predecessor and remain the canonical empirical reference.

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